Ocean Engineering 286 (2023) 115596

Contents lists available at ScienceDirect

Ocean Engineering

journal homepage: www.elsevier.com/locate/oceaneng

Estimation of breaking wave region based on coupled wave and storm surge simulations of historical typhoons: A case study of Hangzhou Bay

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ARTICLE INFO

Handling Editor: Prof. A.I. Incecik

Keywords:
Breaking wave region
Breaking criteria
Univariate probability model
Bivariate probability model
Environmental contour

ABSTRACT

Breaking waves pose a serious threat to offshore structures. Estimating the breaking wave region is hence important for structural design and construction. This paper developed a numerical approach for estimating the breaking wave region based on the coupled wave and storm surge simulations of typhoons. The SWAN + ADCIRC coupling model was employed to hindcast the wave height, wave period, and storm surge under historical typhoons. The values of two breaking prediction variables at different mean return periods (MRPs) were calculated by probability models. The breaking wave region can then be estimated by either a univariate or bivariate probability model of the breaking prediction variables. The developed approach was illustrated using Hangzhou Bay as a case study. Hindcast simulations of 40 typhoons in the past 25 years were conducted. The effects of breaking criteria, probability model, and breaking threshold on the breaking wave region were investigated and discussed as a function of MRP. The main findings include that 1) the waves on the north shore of Hangzhou Bay break more easily; 2) the distribution and area of the estimated breaking wave region depend on the selection of breaking criteria and thresholds; and 3) neglecting the correlation between variables results in an underestimation of breaking wave hazards. This study provides engineers with an efficient tool to estimate the breaking wave hazards for projects located in tropical cyclone-prone sea areas.

1. Introduction

The offshore infrastructures, including sea-crossing bridges and offshore wind farms that locate close to the coast, are at a high risk of being threatened by breaking waves, especially during disastrous weather events such as tropical cyclones (Padgett et al., 2012; Huang and Xiao, 2009; Hong et al., 2021; Hallowell et al., 2021; Wei et al., 2023). The breaking wave is a type of extreme wave that occurs in nearshore waters under strong wind, swell, and storm surge. Cuomo et al. (2010) have noted that wave impact loads of breaking waves could be 10–50 times greater than those of non-breaking waves. The structural vibration excited by breaking waves (Yu et al., 1997), can amplify the structural dynamic response (Choi et al., 2015), shorten the structural fatigue life (Li et al., 2001; Wang et al., 1996), and even lead to the collapse of bridge structures (Fang et al., 2019). Therefore, it is important to estimate breaking wave regions for operational and structural safety, especially when a sea-crossing bridge is located on a site with breaking wave hazards (Wei et al., 2022a,b). Estimating breaking wave regions is challenging since the breaking waves are site-specific (Wei et al., 2014), and depend on the intensities of typhoon wind, wave, and storm surge conditions (Zhang et al., 2020). Therefore, the objective of this study is to develop an efficient numerical approach to assess the location and breaking wave regions in a given site under typhoons. The developed approach employs not only the numerical metocean simulations of historical typhoons but also the breaking criteria and hazard probability models, to estimate the breaking wave region as a function of the mean return period (MRP).

During the process of a wave propagating from the offshore to the nearshore zone, the large wave may break. Many theoretical studies have investigated whether and when a wave can break. Various complex effects, such as shoaling, refraction, wave-wave interaction, wave steepness, relative wave height, slope, etc., can break the wave (Wei et al., 2022a,b; Aggarwal et al., 2020). Many breaking criteria have been proposed to estimate the breaking wave conditions. The first breaking criterion was proposed by McCowan (1894), who defines that wave breaks when the ratio of wave height over water depth exceeds 0.78. Goda (1975) proposed an empirical formula that predicted the breaking condition by wave steepness, including the effect of seafloor slope.

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https://doi.org/10.1016/j.oceaneng.2023.115596
Received 14 February 2023; Received in revised form 18 April 2023; Accepted 10 August 2023
Available online 14 August 2023

Kamphuis (1991) formulated the breaking limit in terms of significant wave height rather than the wave height of an individual wave. Battjes and Groenendijk (2000) proposed a simplified breaking limit of wave steepness as a function of water depth and breaking wavelength. Stringari and Power (2019) presented a machine learning method using deep learning techniques to predict breaking waves. Hallowell et al. (2021) compared different breaking criteria with given sea state data and concluded that the breaking wave hazard estimation was highly sensitive to the breaking criteria chosen.

The estimations of the breaking wave region can also be conducted by field measurements. Melville and Matusov (2002) used aerial imaging to measure breaking waves and provided a statistical description of breaking wave processes. Almeida et al. (2013) applied Terrestrial-mounted Light Detection and Ranging (LIDAR) to quantify wave runup and foreshore changes. However, the field measurements of typhoon-induced metocean conditions are not always sufficient for every location of offshore structures due to a lack of instrumentation, low signal-to-noise ratio, and poor operational reliability during typhoons. Therefore, numerical metocean models were applied to simulate typhoon-induced metocean conditions as a supplement to field measurements. Ma et al. (2012) predicted the breaking wave by the Non-Hydrostatic Wave (NHWAVE) model and concluded that the breaking wave regions can be predicted by including shoaling, diffraction, refraction, wave-wave interaction, wave-current interaction, and wave runup effects in the simulation. However, the numerical simulation of waves, storm surges, and other metocean conditions has not been applied in the estimation of breaking waves under typhoons.

To address this issue, an efficient numerical approach that combines the coupled wave and storm surge model with the probability methods is developed to estimate breaking wave regions. The rest of this paper is organized as follows: first, the framework of the developed approach is described, including the typhoon wind field, coupled theory of wave and storm surge simulation models, breaking prediction variables, probability models, and estimation of the breaking wave region; second, the approach is illustrated using Hangzhou Bay as a case study under 40 historical typhoons; third, the effects of breaking criteria, probability types and breaking thresholds on the distribution and area of breaking wave regions are investigated.

2. Framework of the approach

This section describes the framework of a numerical approach for estimating the breaking wave region. The developed approach employs the fully coupled model, the Simulating WAves Nearshore (SWAN) model, and the ADvanced CIRCulation (ADCIRC) model (Xie et al., 2016) to simulate the wave height, wave period, and storm surge under typhoons. Breaking wave criteria and probability models of producing breaking wave prediction variables at a given MRP are then combined to estimate the breaking wave region. The overall procedure of the developed approach mainly includes 1) modeling the wind field of the historical typhoon; 2) simulating metocean conditions under typhoons through SWAN + ADCIRC coupled simulation; 3) building the probability model of producing the breaking prediction variables at different MRPs; and 4) predicting breaking wave conditions using the simulated metocean conditions.

2.1. Wind field of historical typhoons

The fully coupled model, named the SWAN + ADCIRC model, is used to simulate the metocean conditions under historical typhoons. In the SWAN + ADCIRC model, the atmospheric pressure and wind kinematics of typhoons are the main driving forces to simulate wave, current, and surge conditions.

The typhoon wind field is built using a model that comprises the superposition of the parameter wind field (including the moving wind field and gradient wind field) and background wind field. The moving wind field adopts the Miyazaki model (Miyazaki et al., 1962), and can be calculated according to the change in the position of the typhoon center as follows:

$$v_m = v_{mc}\exp\left(-\frac{r}{500000}\right) \tag{1}$$

where v_m is the moving wind speed of the calculation point in the wind field, v_{mc} is the moving wind speed at the typhoon's center, and r is the distance from the calculation point to the typhoon center.

The gradient wind field adopts the Holland pressure model. The gradient wind speed v_g is obtained by solving the gradient equation that includes the centrifugal force, Coriolis force, and pressure gradient force for circulating air particles (ignoring the influence of ground friction) and is expressed as follows:

$$v_g = \sqrt{v_{\max}^2 e^{\left[1-(R/r)^B\right]}(R/r)^B + \left(\frac{rf}{2}\right)^2} - \frac{rf}{2} \tag{2}$$

where f is the Coriolis force parameter $f = 2\omega \sin\varphi$, ω is the angular velocity of the Earth's rotation, φ is the latitude, and B is the pressure profile parameter, which can be solved by using the maximum gradient wind speed v_{max} in Eq. (3). v_{max} approximates the maximum sustained wind speed obtained from the China Meteorological Administration (Wei et al., 2019), and R is the maximum wind speed radius, calculated using Eq. (4) proposed by Zhou et al. (2018):

$$B \approx \frac{v_{\max}^2 \rho_a e}{\Delta P} \tag{3}$$

$$R = 29.178\exp[0.0158(P_0 - 900)] \tag{4}$$

where P_0 is the central pressure of the typhoon; ρ_0 is the air density; and ΔP is the difference between the peripheral air pressure (set as 1010 hPa) and the central air pressure P_0.

The gradient wind v_g is converted from the atmospheric boundary layer to sea level by using a height coefficient of 0.9. The gradient wind at sea level v_{gs} is superimposed with the moving wind vector v_m at sea level required to form an empirical typhoon field v_e, as shown in Eq. (5):

$$v_e = v_{gs} + v_m \tag{5}$$

It should be noted that the empirical typhoon field is not always valid for wind that is far away from the typhoon eye. The background wind field can simulate the wind field far from the typhoon center more accurately than the parameter wind field (Murty et al., 2020). Therefore, the global-scale background wind field provided by the Cross-Calibrated Multi-Platform (CCMP) reanalysis database is used and superimposed on the above wind field to improve the parametric typhoon wind field. According to the proportion of distance (Pan et al., 2016), the transition region wind v_{sup} is composed of a combination of empirical typhoon v_e and CCMP background wind v_b through superposition of the coefficient α related to the distance to the typhoon's center, as shown in Eq. (6).

$$v_{\text{sup}} = \begin{cases} v_e \; r < R_1 \\ (1-\alpha)v_e + \alpha v_b \; R_1 \le r \le R_2 \\ v_b \; r > R_2 \end{cases} \tag{6}$$

in which the parameter α can be calculated by Eq. (7):

$$\alpha = \frac{r - R_1}{R_2 - R_1} \tag{7}$$

in which the value of R_1 and R_2 is set to be 300 km and 400 km, respectively.

2.2. Coupled wave and storm surge model

SWAN can compute the wave conditions by the two-dimensional wave energy spectrum in the frequency and direction domains, which considers physical processes such as wind energy input, white hat dissipation, bottom friction, and nonlinear wave-wave interactions. The governing equation in spherical coordinates is expressed as follows:

$$\frac{\partial N}{\partial t}+\frac{\partial c_{\lambda}N}{\partial \lambda}+\cos^{-1}\varphi\frac{\partial c_{\varphi}\cos\varphi N}{\partial \varphi}+\frac{\partial c^{\sigma}N}{\partial \sigma}+\frac{\partial c^{\theta}N}{\partial \theta}=\frac{S_{tot}}{\sigma} \tag{8}$$

where σ is the relative radian or circular frequency; θ is the wave propagation direction; λ and φ are longitude and latitude, respectively; and c_{λ} and c_{φ} are the velocities of the wave energy propagation in the longitudinal and latitudinal directions, respectively. c_{σ} and c_{θ} are the wave energy propagation velocities in spectral space; S_{tot} is the source or sink term that represents the effects of generation, dissipation of energy, and nonlinear wave-wave interactions; and N is the action density, which is defined as:

$$N(\lambda,\varphi,\sigma,\theta)=\frac{E(\lambda,\varphi,\sigma,\theta)}{\sigma} \tag{9}$$

in which E represents the wave energy density. The source term includes the input energy from the wind, dissipation due to the bottom friction, breaking wave, and nonlinear wave-wave interactions.

The ADCIRC model is developed based on the generalized wave continuity equation and momentum equation for solving the velocity and water level. The model can resolve complex geometry and bathymetry using an unstructured triangular grid. The governing equations in spherical coordinates are expressed as follows:

$$\frac{\partial \zeta}{\partial t}+\frac{1}{R\cos\varphi}\left[\frac{\partial(UH)}{\partial \lambda}+\frac{\partial(VH\cos\varphi)}{\partial \varphi}\right]=0 \tag{10}$$

$$\frac{\partial U}{\partial t}+\frac{1}{R\cos\varphi}U\frac{\partial U}{\partial \lambda}+\frac{V}{R}\frac{\partial U}{\partial \varphi}-\left(\frac{\tan\varphi}{R}U+f\right)V=-\frac{1}{R\cos\varphi}\frac{\partial}{\partial \lambda}\left[\frac{p_s}{\rho_0}+g(\zeta-\alpha\eta)\right]$$
$$+\frac{\nu_T}{H}\frac{\partial}{\partial \lambda}\left[\frac{\partial(UH)}{\partial \lambda}+\frac{\partial(VH)}{\partial \varphi}\right]+$$
$$\frac{\tau_{s\lambda}}{\rho_0 H}-\tau*U \tag{11}$$

$$\frac{\partial V}{\partial t}+\frac{1}{R\cos\varphi}U\frac{\partial V}{\partial \lambda}+\frac{V}{R}\frac{\partial V}{\partial \varphi}-\left(\frac{\tan\varphi}{R}U+f\right)U=-\frac{1}{R}\frac{\partial}{\partial \varphi}\left[\frac{p_s}{\rho_0}+g(\zeta-\alpha\eta)\right]$$
$$+\frac{\nu_T}{H}\frac{\partial}{\partial \varphi}\left[\frac{\partial(VH)}{\partial \lambda}+\frac{\partial(VH)}{\partial \varphi}\right]+$$
$$\frac{\tau_{s\varphi}}{\rho_0 H}-\tau*V \tag{12}$$

where t is time; ζ is the free surface elevation relative to the geoid; U and V are the depth-integrated velocities in the west-east and south-north directions, respectively; $H=\zeta+h$ is the total water depth; h is the bathymetric water depth relative to the geoid; f is the Coriolis parameter, where Ω is the angular speed of the earth and $f = 2\Omega \sin\varphi$; p_s is the atmospheric pressure; R is the radius of the Earth; g is the gravitational acceleration; ρ_0 is the reference density of water; η is the Newtonian equilibrium tide potential; α is the effective earth elasticity factor; ν_T is the depth-averaged horizontal eddy viscosity coefficient; and $\tau_{S\lambda}$ and $\tau_{S\varphi}$ are the surface wind stresses in the longitudinal and latitudinal directions, respectively. They can be computed by a standard quadratic air-sea drag law. The bottom friction term τ_* is defined as follows:

$$\tau_*=\frac{C_f\left(U^2+V^2\right)^{1/2}}{H} \tag{13}$$

where C_f is the bottom friction coefficient.

SWAN and ADCIRC use the same set of grids and are driven by the typhoon wind field and boundary water level. The coupling calculation between these models is achieved by ADCIRC's calculation and transfer of the calculated velocity and water level to SWAN, which then computes and sends the wave radiation stress back to ADCIRC for the next step. The coupling details can be found in the literature (Dietrich et al., 2011).

2.3. Breaking prediction variables

To estimate the breaking wave, the study employs four common breaking criteria, namely, the McCowan (1894), Battjes and Groenendijk (2000), Goda (1975), and Kamphuis (1991) criteria, each of which is characterized by a formula that accounts for the different dominating factors. The formulas and dominating factors for the above breaking criteria are given in Table 1.

In these criteria, H_b defines the breaking limit of the wave height. d is the actual water depth and equals $d = d_0 + d_s$, in which d_0 is the mean water depth and d_s is the storm surge, which is the rise in the sea level caused by a tropical cyclone. L_b is the breaking wavelength and can be used to calculate the period according to the linear dispersion relation.

The data obtained from the numerical simulation are the significant wave height H_s, the mean period T_m, the mean water depth d_0, and the storm surge d_s. The maximum wave height H_{max} and T_m are used in the McCowan, Battjes, and Goda criteria. The wave height in the Kamphuis criterion adopts the significant wave height H_s, and the period is the spectral peak cycle T_p. Therefore, it is necessary to convert H_s in the numerical simulation into H_{max} in McCowan, Battjes, and Goda criteria and convert T_m into T_p in the Kamphuis criterion. Since the case study is illustrated using Hangzhou Bay on China's eastern coast, the following functions recommended by Chinese code (JTS145-2015) are used to convert the data:

$$T_p=1.21T_m \tag{14}$$

$$H_{\max}=1.52H_s \tag{15}$$

According to Hallowell et al. (2021), the breaking wave can be estimated using two normalized breaking prediction variables x_1 and x_2, which are defined by Eq. (16) and Eq. (17), respectively:

$$x_1=\frac{H_{\max}}{d} \tag{16}$$

$$x_2=\frac{T_z^2 g}{d} \tag{17}$$

It should be noted that in the McCowan, Battjes, and Goda criteria T_z is the average period T_m, while T_z should be equal to T_p in the Kamphuis criterion.

Fig. 1 shows four breaking criteria in the coordinates of x_1 and x_2. The McCowan criterion is only related to x_1, and the curve is perpendicular to the x_1-axis. The curve of the Kamphuis criterion is the steepest one, followed by the Goda criterion. The Battjes criterion curve changes most gently compared with the Kamphuis and Goda criteria.

2.4. Probability model of breaking prediction variable

Considering that the record of historical typhoons is usually less than

Table 1
List of breaking criteria.

Criteria	Formula	Dominating factor
McCowan	$H_b = 0.78d$	depth
Battjes	$H_b = 0.142L_b \tanh\left(\frac{0.8}{0.88}\frac{2\pi d}{L_b}\right)$	steepness
Goda	$H_b = 0.17L_b\left\{1-\exp\left[-1.5\frac{\pi d}{L_b}\left(1+15\tan(\beta)^{\frac{4}{3}}\right)\right]\right\}$	steepness and slope
Kamphuis	$H_b = 0.095\exp[4\tan(\beta)]L_b\tanh\left(\frac{2\pi d}{L_b}\right)$	steepness and slope

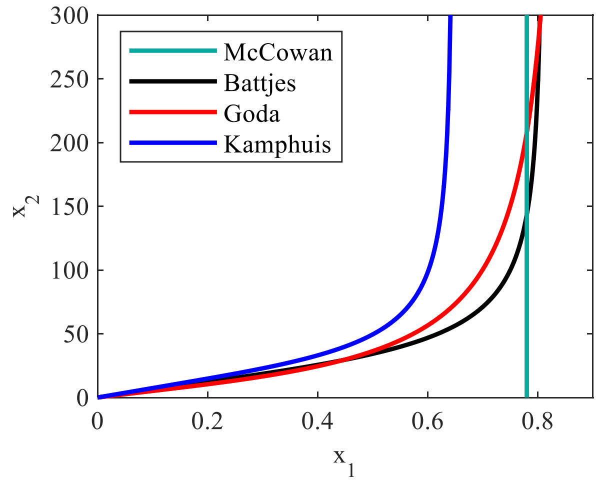

Fig. 1. Breaking limit curves of different breaking wave criteria.

50 years in China, it is necessary to extrapolate the data by the probability method for the estimation of the environmental conditions with an MRP longer than 50 years. In this study, two types of probability models are employed in estimating the breaking wave, depending on the consideration or neglect of the correlation between the variables x_1 and x_2.

2.4.1. Type I: Univariate probability model

When the correlation is neglected, the variables are considered independent. The marginal distribution of x_1 and x_2 must be calculated based on their data before the values of x_1 and x_2 at a given MRP can be calculated as follows:

$$T = \frac{L_t}{m(1 - F(x))} \tag{18}$$

where $F(x)$ is the distribution function of variable x, m is the number of events generating variable x, and L_t is the time length of the statistical variable x.

2.4.2. Type II: Bivariate probability model

If the correlation is considered, the copula function can construct the bivariate probability model between the marginal distributions of x_1 and x_2. According to Sklar's theory (Sklar, 1959), there are m random variables $x_1, x_2, x_3, \ldots, x_m$. For a marginal distribution of $F_1(x_1), F_2(x_2), F_3(x_3), \ldots, F_m(x_m)$, then there exists a copula function satisfying:

$$F(x_1, x_2, \cdots, x_m) = C(F_1(x_1), F_2(x_2), \cdots, F_m(x_m)) \tag{19}$$

where $F(x_1, x_2, \ldots, x_m)$ is the joint distribution function of m random variables and its corresponding joint probability density function $f(x_1, x_2, \ldots, x_m)$ is given by:

$$f(x_1, x_2, \cdots, x_m) = \frac{\partial^m F(x_1, x_2, \cdots, x_m)}{\partial x_1 \partial x_2, \cdots, \partial x_m} \tag{20}$$

The inverse first-order reliability method (IFORM) is an indispensable tool for obtaining environmental contours, which is a probabilistic representation of the frequency and magnitude of extreme environmental events over a given period. This process involves converting the equal probability circle, which possesses a constant radius β on the uncorrelated standard normal plane, to the plane of physical joint random variables. In the uncorrelated standard normal space, the radius of the equal probability circle β equals the target reliability index. The equal probability circle radius β with a T-year MRP can be calculated by using the following formula:

$$\beta = \Phi^{-1}\left(1 - \frac{1}{N \cdot T}\right) \tag{21}$$

where Φ^{-1} is the inverse cumulative distribution function of the standard normal distribution, and N is the annual incidence of random events. According to the principle of equal probability, namely, the Rosenblatt transformation, the transformation relationship between the 2D standard normal space and physical joint random variable space can be expressed as follows:

$$\varphi(u_1) = F_1(x_1) \tag{22}$$

$$\varphi(u_2) = F_{2|1}(x_2|x_1) \tag{23}$$

where u_1 and u_2 are two uncorrelated standard normal random variables and F_1 and $F_{2|1}$ are the distribution of variable x_1 and the conditional distribution of x_2 under variable x_1, respectively.

2.5. Estimation of breaking wave region

For a given map, the estimation of breaking waves can be categorized into two types depending on the type of probability models of x_1 and x_2. When the univariate model is used, there is only one data point for the given MRP, as illustrated in Fig. 2 (a). To determine if the wave is breaking, a selected breaking criterion is applied. If the point falls to the left of the breaking limit curve, the wave is considered non-breaking; otherwise, it is a breaking wave.

When the bivariate model is used, there is a contour for the given MRP, as shown in Fig. 2 (b). In this case, the breaking limit cuts across the contour, indicating that some wave conditions are breaking while others are not. Since estimating the breaking using the same approach as the univariate model is difficult, the breaking ratio of the contour β_b is defined as Eq. (24) to estimate the breaking wave by the percentage of breaking over the overall contour:

$$\beta_b = \frac{L_{breaking}}{L_{contour}} \tag{24}$$

in which $L_{breaking}$ defines the arc length of the contour on the right of the breaking limit, and $L_{contour}$ is the overall length of the contour. To ensure the safety of the structure, the threshold of the breaking ratio should be set according to the safety design level of the structure. When the ratio surpasses the threshold, it means that the wave at the node is breaking. It should be noted that the threshold can be set to zero, which is the strictest situation in which the wave at such MRP is considered breaking once there are any points beyond the breaking limit. The breaking wave region can be obtained after the wave conditions are checked at all the nodes by the procedure listed above.

3. Example site and numerical model

3.1. Description and modeling of Hangzhou Bay

Hangzhou Bay, which is a typhoon-prone area, is taken as the example site. The example site is economically developed and densely populated, whereas many sea-crossing bridges have been or are being built, as depicted in Fig. 3.

The site model encompasses a range from 115° E to 132° E in longitude and from 25° N to 41° N in latitude, covering parts of the Yellow Sea and the East China Sea. An unstructured triangular grid is adopted to discretize the site model with 70,144 grids and 37,064 nodes, as shown in Fig. 4. The grid size ranges from as small as 150 m in the Hangzhou Bay region to as large as 60,000 m in the offshore sea area. Such a transition of grid size results in a good balance between computational accuracy and convergence efficiency.

The boundary water level is simulated by the TPXO9_atlas Atlas mode by applying eight primary tidal components (M2, S2, N2, K2, K1,

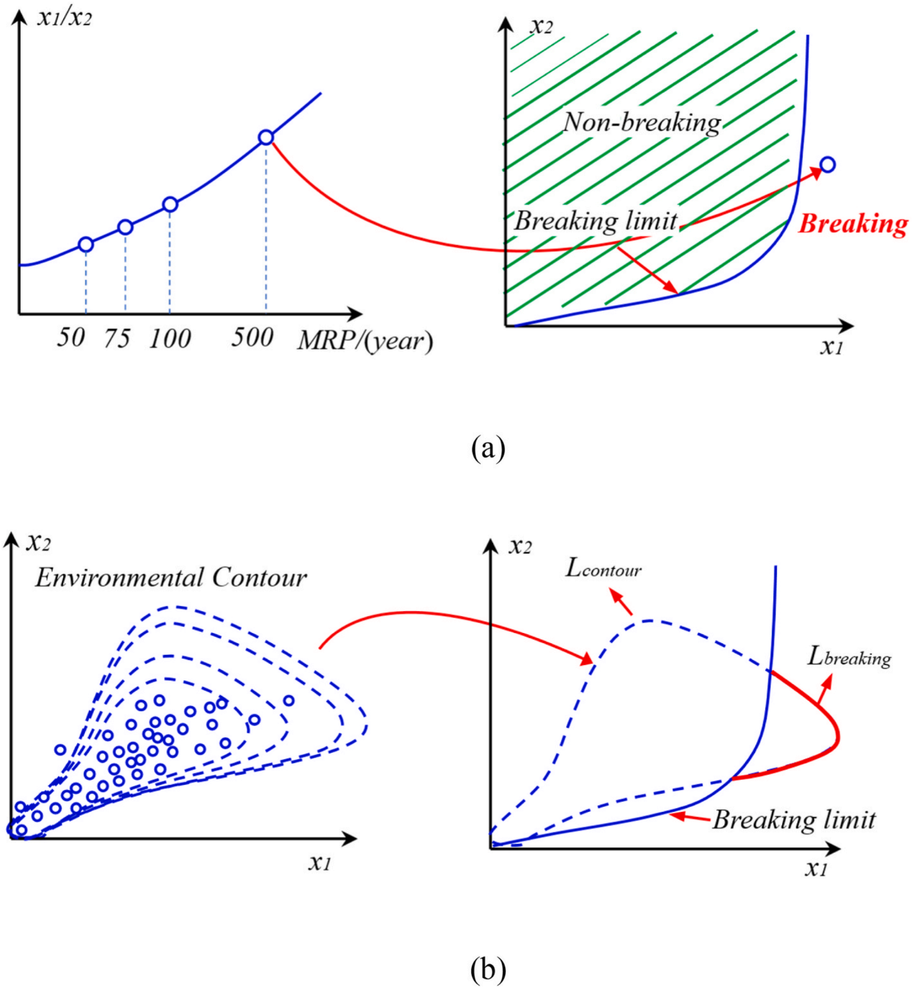

Fig. 2. Breaking wave prediction under univariate and bivariate probability models: (a) univariate probability model and (b) bivariate probability model.

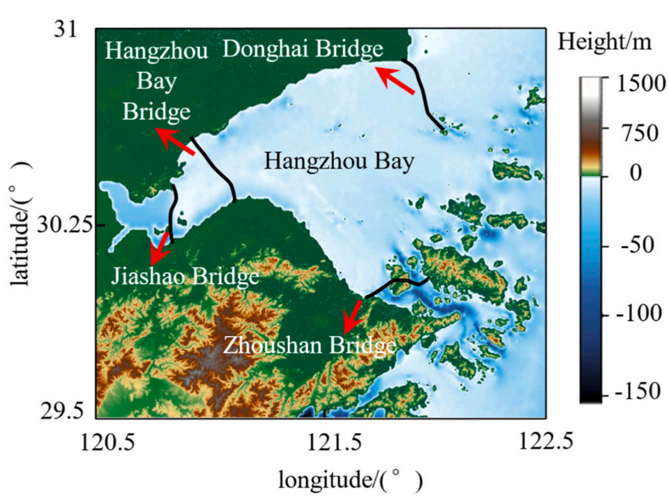

Fig. 3. Elevation map of Hangzhou Bay and locations of sea-crossing bridges.

O1, P1, and Q1). The time histories for significant wave height H_s, the mean period T_m, the mean water depth d_0, and storm surge d_s over the whole track of the typhoon can then be computed and output for all the nodes in the region of interest.

3.2. Numerical simulation and validation of historical typhoons

The historical typhoon data for the example site are downloaded from the websites of China Typhoon Online (http://www.typhoon.org.cn) and China National Meteorological Center (http://typhoon.nmc.cn). They provide the tropical cyclone center's location, the central pressure, translation speed, heading direction, and the maximum sustained wind speed of the historical typhoons. A total of 40 historical typhoons from 1987 to 2018 that went through the circle of Hangzhou Bay, with a circle center coordinate of 121.8°N, 30.4°E, and a radius of 250 km, are selected, as shown in Fig. 5.

The effectiveness of the numerical simulation is then validated by comparing the simulated data with the measured data of the significant wave height H_s and storm surge d_s. The significant wave height H_s for Typhoon Winnie from 0:00 on August 17, 1997, to 0:00 on August 20, 1997, is analyzed. As shown in Fig. 6 (a), the measured maximum wave height H_s at the Zhenhai station is 2.97 m, while the simulated maximum wave height is 2.98 m, resulting in a relative error in the wave height of 0.3%. Fig. 6 (b) compares the simulated and measured significant wave heights H_s of Typhoon No. 1509 Canhong from 12:00 on July 9, 2015, to 12:00 on July 11, 2015, at Nanji Station. According to Fig. 6 (b), the measured maximum significant wave height is 5.69 m, and the simulated maximum significant wave height is 5.74 m, with a relative error of 0.9%.

The evaluation of the effectiveness of numerical models in simulating the storm surge d_s at Dinghai station during Typhoon Winnie, which occurred from 0:00 on August 17, 1997, to 0:00 on August 20, 1997. As

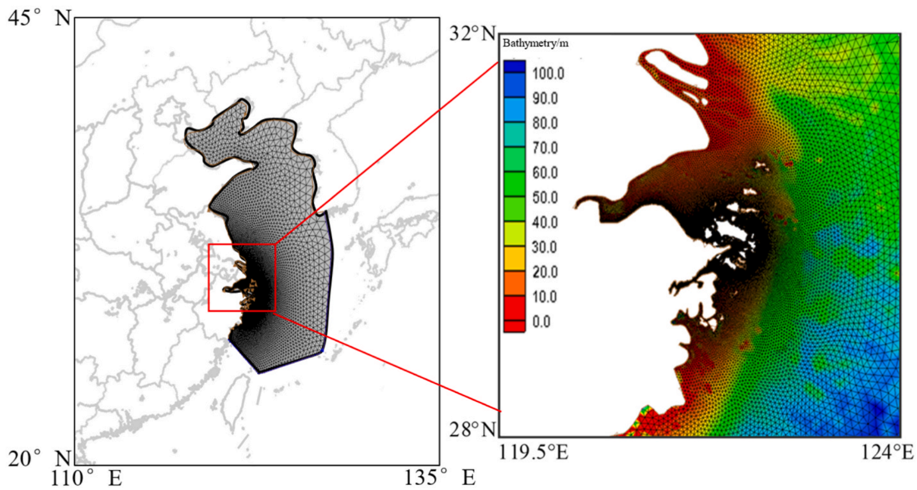

Fig. 4. Model domain with unstructured triangular grids in SWAN.

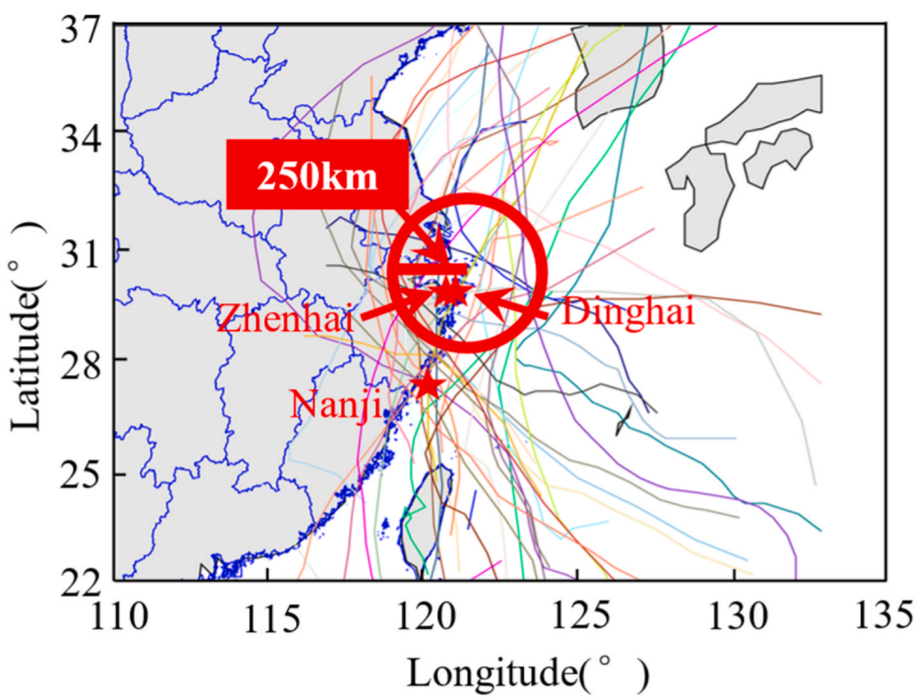

Fig. 5. Traces for selected tropical cyclones.

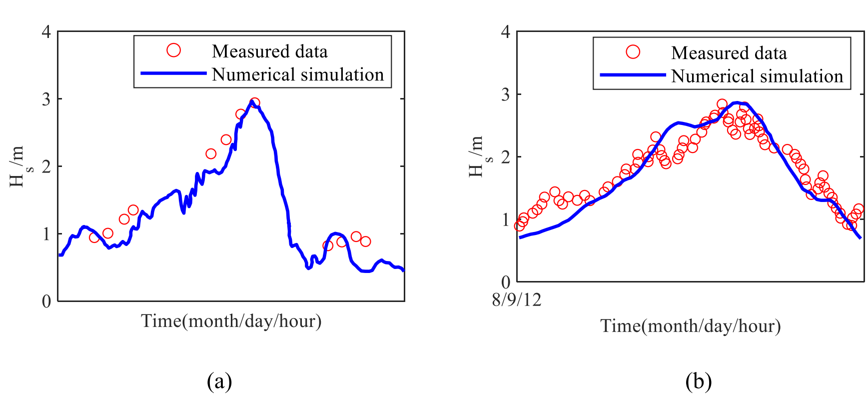

Fig. 6. Comparison between the simulated and measured significant wave heights: (a) Zhenhai station; and (b) Nanji station.

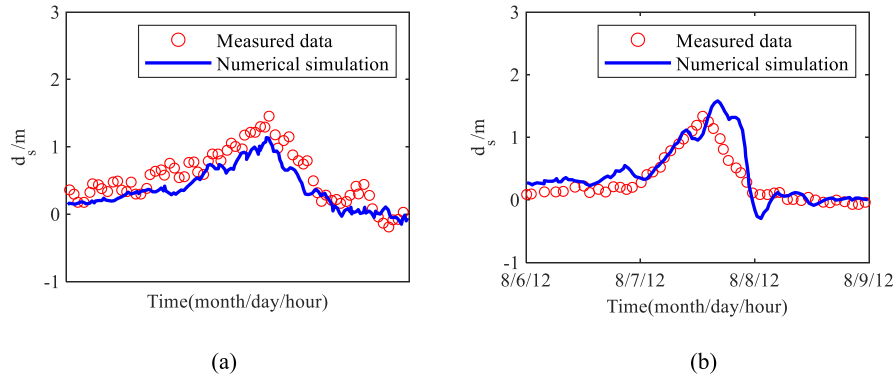

Fig. 7. Comparison between the simulated and measured storm surges: (a) Dinghai station; and (b) "MJ" station.

shown in Fig. 7 (a), the maximum d_s is measured to be 1.44 m, and the simulated maximum d_s is 1.17 m. Fig. 7 (b) compares the numerical simulation and measured storm surge d_s at the "MJ" station during Typhoon Anemone, which occurred from 0:00 on August 6, 2012, to 0:00 on August 9, 2012. The measured maximum d_s is 1.31 m, and the simulated maximum d_s is 1.58 m. The satisfactory agreement between the simulated and measured H_s and d_s values attests to the credibility and soundness of the simulation modeling.

3.3. Selection of breaking prediction variables x_1 and x_2

To estimate the breaking wave region, the wave and storm surge data at each node of the numerical model should be output first. For each node in the example site, the time histories of x_1 and x_2 are calculated according to Eqs. (16) and (17).

Since the wave data obtained by numerical simulation are time-history data, it is necessary to determine which moment of data to choose for the determination of breaking waves. Fig. 8 shows the time histories of x_1 and x_2 at 121.5° E and 29.5° N during Typhoon Winnie. The data represented by the red circle and the blue circle are the values of x_1 and x_2, respectively. The waves break at the peak times of x_1 and x_2 following Kamphuis' criteria. It can be seen that the peak moments of x_1 and x_2 are not the same but relatively close. Considering that the time difference between the two peak moments is a mere 20 min, it is assumed to be negligible in the estimation of the breaking waves, which utilizes the simultaneously obtained peak values of x_1 and x_2.

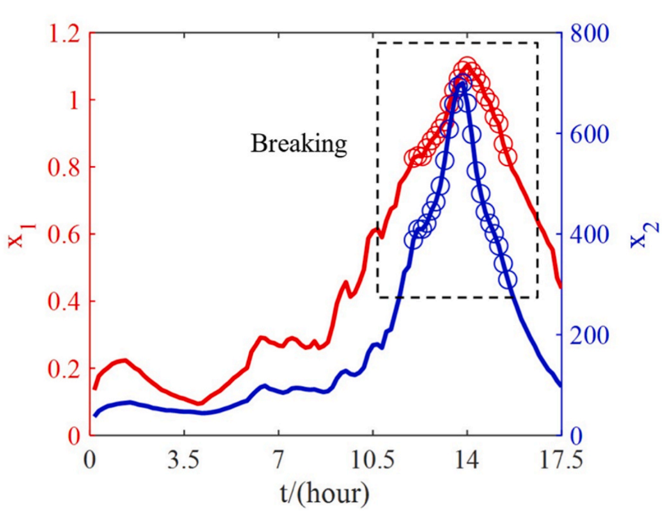

Fig. 8. Time history of x_1 and x_2 at 121.5° E and 29.5° N during Typhoon Winnie.

3.4. Probability modeling for breaking wave variables

In the univariate probability model, the data are fit using some common distribution functions. Taking the simulated data at 122.0°E and 30°N as an example, the distributions of variables x_1 and x_2 are fitted by Rayleigh, Weibull, GEV, and logistic distributions, respectively. The root mean square error (RMSE) is employed to evaluate the quality of fitting, which focuses on the minimization of residuals between the fitted and the empirical distribution:

$$RMSE = \sqrt{\frac{1}{N_0}\sum_{i=1}^{N_0}(p_i - \widehat{p}_i)^2} \tag{25}$$

where N_0 is the number of data points and p_i and $\widehat{p}_i$ are the empirical frequency and theoretical frequency, respectively. Fig. 9 compares the fitted cumulative distribution function (CDF) with the empirical functions of x_1 and x_2. The x_1 and x_2 distributions obey the Weibull distribution, of which the RMSEs are 0.0163 and 0.0147, respectively. The probability density function (PDF) of the Weibull distribution can be described as:

$$f(x) = \frac{b}{a}\left(\frac{x}{a}\right)^{b-1}\exp\left(-\left(\frac{x}{a}\right)^{b}\right) \tag{26}$$

where a and b are the scale and shape parameters of the Weibull distribution, respectively. The parameters fitted for the Weibull distribution are $a = 0.0617$ and b = 1.2534 for x_1 and $a = 12.6$ and b = 1.8084 for x_2.

The choice of copula function is important to establish the bivariate probability model. Three common bivariate Archimedean copula functions, including the Frank, Clayton, and Gumbel copulas, are used to fit the pairs of variables x_1 and x_2. Their probability distribution and density function are expressed as follows:

Frank copula:

$$C_\theta(u,v) = -\frac{1}{\theta}\ln\left(1+\frac{(e^{-\theta u}-1)(e^{-\theta v}-1)}{(e^{-\theta}-1)}\right) \tag{27}$$

$$c_\theta(u,v) = \frac{\partial^2 C_\theta(u,v)}{\partial u \partial v} = \frac{-\theta(e^{-\theta}-1)e^{-\theta(u+v)}}{\left[(e^{-\theta}-1)+(e^{-\theta u}-1)(e^{-\theta v}-1)\right]^2} \tag{28}$$

Clayton copula:

$$C_\theta(u,v) = \left(\left[u^\theta + v^\theta\right]-1\right)^{-1/\theta} \tag{29}$$

$$c_\theta(u,v) = \frac{\partial^2 C_\theta(u,v)}{\partial u \partial v} = (1+\theta)(uv)^{-\theta-1}\left(u^{-\theta}+v^{-\theta}-1\right)^{-2-1/\theta} \tag{30}$$

Gumbel copula:

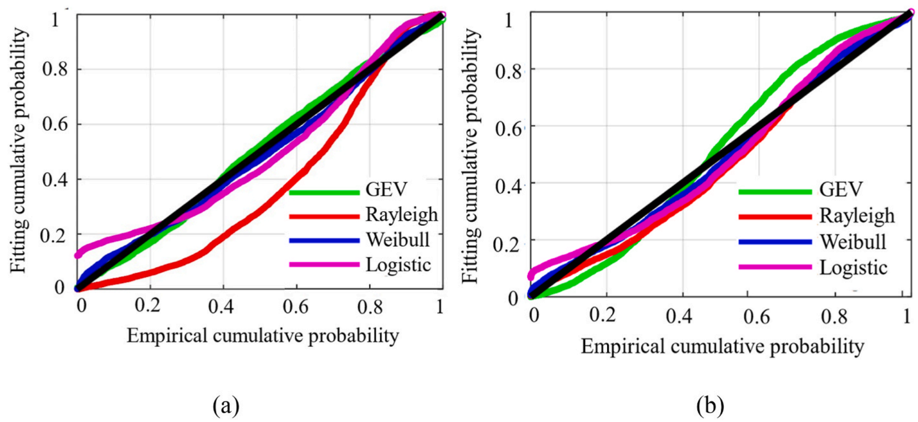

Fig. 9. The fitted CDF versus the empirical CDF: (a) x_1; and (b) x_2.

$$C_\theta(u,v) = \exp\left(- \left[(-\ln u)^\theta + (-\ln v)^\theta \right]^{1/\theta} \right) \tag{31}$$

$$\begin{aligned} c_\theta(u,v) &= \frac{\partial^2 C_\theta(u,v)}{\partial u \partial v} \\ &= \frac{C(u,v;\theta)(\ln u \times \ln v)^{\theta-1}}{uv\left[(-\ln u)^\theta + (-\ln v)^\theta\right]^{2-1/\theta}} \left\{ \left[(-\ln u)^\theta + (-\ln v)^\theta\right] + \theta - 1 \right\} \end{aligned} \tag{32}$$

where θ is the parameter of the copula function, which can be estimated using the maximum likelihood estimate method.

Fig. 10 shows the bivariate joint distribution of x_1 and x_2 with the copula function, and the fitted and empirical CDF of the variables are illustrated in Fig. 11. Compared with the results of Clayton and Gumbel, the RMSE value fitted by the Frank function is the minimum. Therefore, the joint distributions of x_1 and x_2 are built by the Frank copula.

4. Results and discussion

This section investigates the effects of breaking criteria, probability model, and breaking threshold on the estimation of breaking wave region as a function of MRP. According to the Port Designer's Handbook: Recommendations and Guidelines (Carl, 2003), there is the following relationship between structural service life n and design mean return period of wave hazard *MRP*:

$$P = 1 - \left(1 - \frac{1}{MRP}\right)^n \tag{33}$$

where P is the probability that the structure will suffer a wave greater than the design wave within the design mean return period. The service life is usually 25–50 years for an offshore wind turbine and 100 years for sea-crossing bridges. Fig. 12 plots the relationship between *MRP* and n, when P is equal to 0.2 and 0.64. When the *MRP* of the wave is 100 years, the probability that the structure will suffer a wave greater than the design wave is as large as 64%. In structural engineering, such loading conditions are mostly considered for checking structural serviceability limit states. However, for the verification of ultimate limit states, the occurrence probability P of the design load should be much smaller than 0.64 (Chen et al., 2019). Assuming P is equal to 0.2, the *MRP* should be 500 years according to Fig. 12. Therefore, four MRPs of breaking wave are selected in the following analyses, including 50 years, 75 years, 100 years, and 500 years.

Four different breaking criteria, two probability models, and different breaking thresholds are employed. The estimated breaking wave regions are presented on a map as demonstrated in Fig. 13. If the breaking wave prediction variables go beyond the breaking limit for a given node in the ocean, the zone is marked blue. The blue zones highlight the spatial distribution of the breaking wave region, which is the sum of all the regions enclosed by the blue nodes.

4.1. Effect of breaking criteria

Fig. 14 illustrates the estimated breaking wave regions using different breaking criteria to investigate the effect of breaking criteria on spatial distribution. The univariate probability model is employed in this case.

Fig. 14 indicates that the Goda criterion results in the broadest distribution of breaking wave regions, while the McCowan criterion generates the smallest region. The breaking wave regions are mainly on the south shore of Hangzhou Bay. The breaking wave regions obtained by

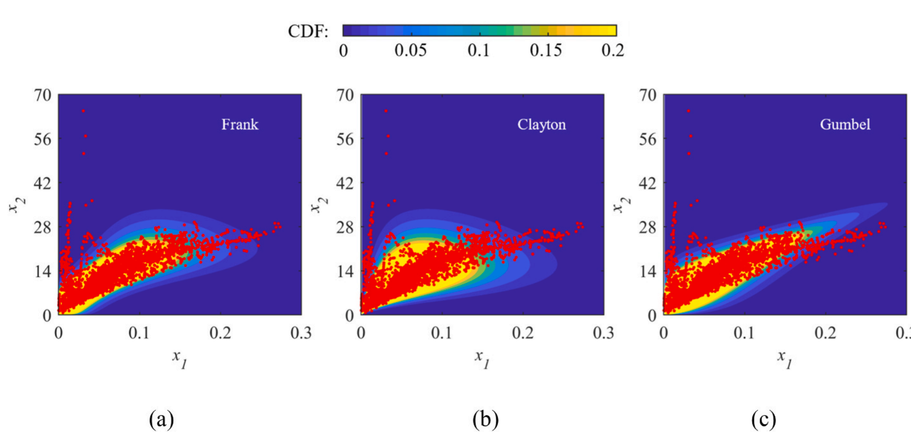

Fig. 10. Different distributions with different copula functions: (a) Frank; (b) Clayton; (c) Gumbel.

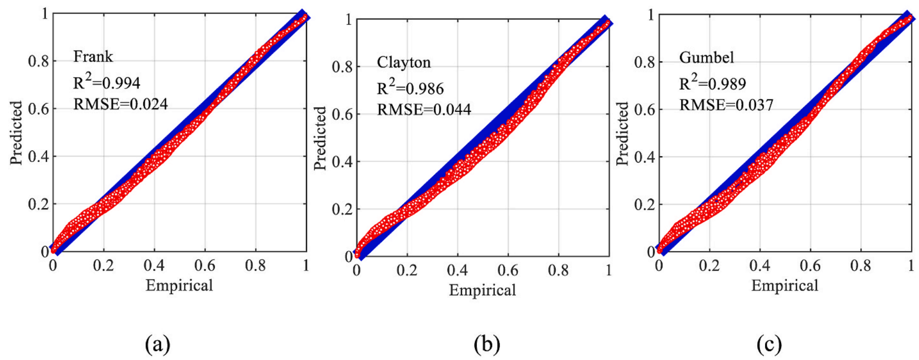

Fig. 11. The predicted CDF versus the empirical CDF: (a) Frank; (b) Clayton; and (c) Gumbel.

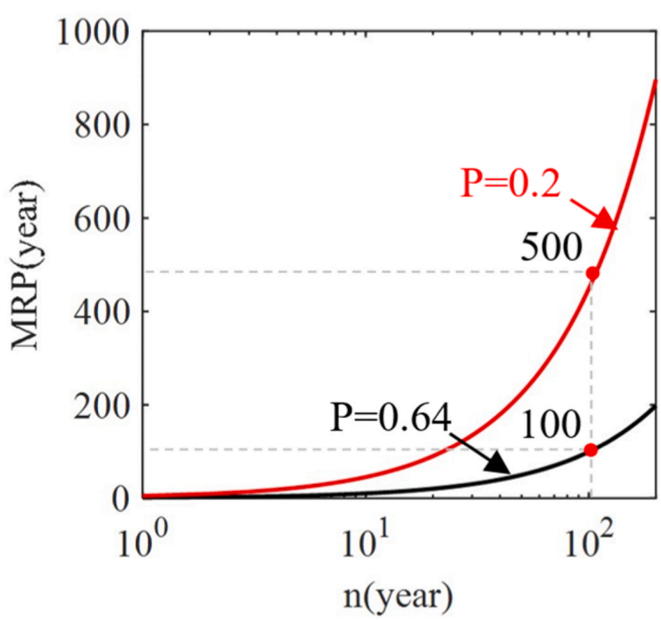

Fig. 12. The relationship between the design mean return period and design service life under different design wave probabilities.

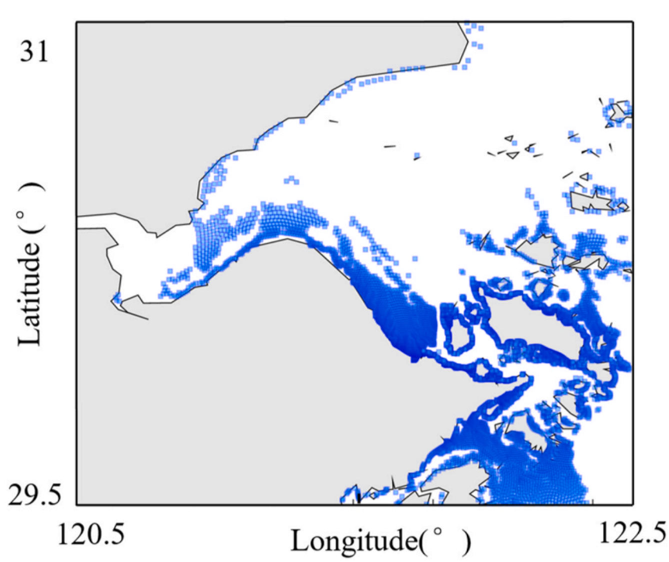

Fig. 13. Spatial distribution of the breaking wave region.

Kamphuis, Battjes, and Goda almost cover the whole sea area of Hangzhou Bay. Most of the bridges in the Hangzhou Bay area are located in the breaking wave regions under the typhoon hazard with an MRP of 500 years regardless of the breaking wave criterion. Except for McCowan's result, all the wind farms will suffer from breaking waves as well.

As shown in Fig. 15, some videos about breaking waves during Typhoon In-Fa in 2021 and Typhoon Muifa in 2022 have been posted through China News (http://www.chinanews.com.cn/m/sh/shipin/cns-d/2021/07-26/news895936.shtml) and TikTok (https://www.douyin.com/video/7143191555279506695).

According to these reports, severe breaking waves were observed around the wind farms located in the middle of Hangzhou Bay. It states that the breaking wave regions estimated by the Kamphuis, Battjes, and Goda criteria have better agreement with the actual events than the McCowan criterion. Moreover, a large number of breaking waves were observed on the south shore of Hangzhou Bay during Typhoon Muifa, as shown in Fig. 15(b). It matches the estimation shown in Fig. 14.

Fig. 16 shows the map of the breaking wave regions as a function of MRP. The breaking wave region extends outward with increasing MRP. Compared with the results obtained by the Kamphuis criterion and Battjes criterion, the breaking wave region obtained by the Goda criterion with an MRP of 75 years and 100 years expands from the nearshore to offshore water on a more extensive scale. When the MRP is 50 years, 75 years, and 100 years, the breaking wave regions are greatly reduced compared to the result with an MRP of 500 years. In addition, the breaking wave regions are mainly distributed near the Zhoushan Sea area and are small on the north shore of Hangzhou Bay.

Fig. 17 compares the area of breaking wave regions with different breaking criteria under different MRPs. The breaking wave area obtained by the McCowan criterion increases slowly with MRP, and the area is approximately 1000 km^2. The average breaking wave area for the other three criteria is approximately 3000 km^2 at an MRP of 50 years, 5000 km^2 at an MRP of 75 years, 7000 km^2 at an MRP of 100 years, and up to 16000 km^2 at an MRP of 500 years. There are slight differences among the results obtained by the Kamphuis, Battjes, and Goda criteria, which are significantly larger than those obtained by the McCowan criterion. The result of the Battjes criterion is much closer to the mean from the Kamphuis, Battjes, and Goda criteria, making it the criterion of choice for the subsequent analyses.

4.2. Effect of the probability model

In this section, the breaking wave region is estimated using the bivariate probability model, in which the breaking threshold of Eq. (24) is set to zero. Fig. 18 exhibits the breaking wave region with an MRP of 100 years estimated by the bivariate probability model. The spatial distribution of the breaking wave region from the bivariate probability model surpasses that from the univariate probability model (Fig. 16 (b)). It mainly increases in the southeast of the Zhoushan Islands. The impact of breaking waves on the Jiashao Bridge and Zhoushan Bridge is

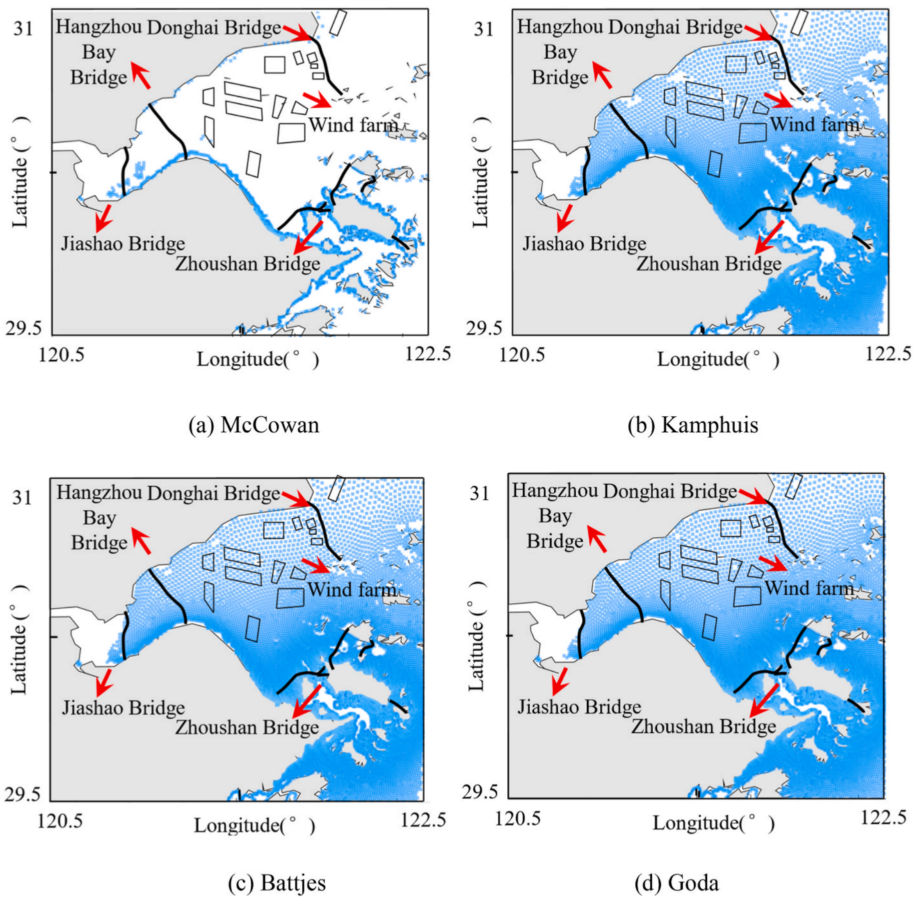

Fig. 14. The spatial distribution of breaking waves with an MRP of 500 years using different breaking criteria (black solid lines indicate the locations of sea-crossing bridges and the quadrilaterals indicate the offshore wind farm in the bay).

Fig. 15. Breaking wave observations: (a) Breaking waves near the Hangzhou Bay Bridge during Typhoon In-Fa; (b) Breaking waves near the Zhoushan Sea area during Typhoon Muifa.

significant, while other bridges and wind farms are less affected.

Fig. 19 compares the breaking wave area using different probability models. The breaking wave area obtained by the univariate probability model under different MRPs is slightly smaller than that of the bivariate probability model. When the MRP becomes longer, their difference can be neglected.

Fig. 20 presents the univariate and bivariate probability models of x_1 and x_2 with the same MRP. The univariate model comprises only one point of data (star point in Fig. 18) at the given MRP, while the bivariate model is a contour (the solid line in Fig. 18). The bivariate model is more rigorous than the univariate model in predicting breaking. As illustrated in Fig. 18, the star point from the univariate model is located at the intersection point of the uppermost and rightmost endpoint tangent of the contour from the bivariate probability model. The dashed line defines the breaking limit. When the star point of the univariate probability model is to the left of the limit, the wave does not break. However, it can be estimated as breaking when the bivariate model is used since the breaking limit passes across the contour. When the breaking threshold is set to zero, it breaks following the procedure developed in Section 2.5.

Although ignoring correlations of the environmental conditions can underestimate the estimated breaking wave area, the univariate probability model is more efficient in the estimation than the bivariate probability model. It can estimate 3000 nodes per hour, while the

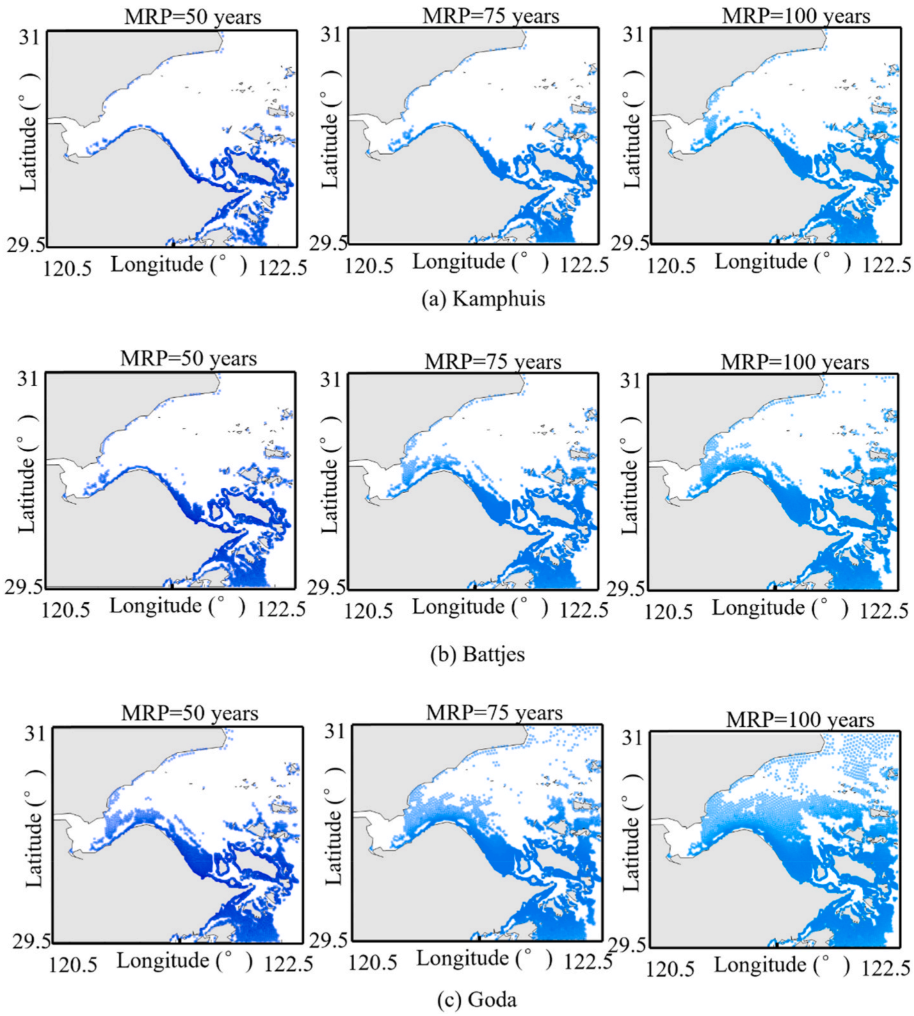

Fig. 16. The spatial distribution of breaking waves with MRPs of 50 years, 75 years, and 100 years using different breaking criteria.

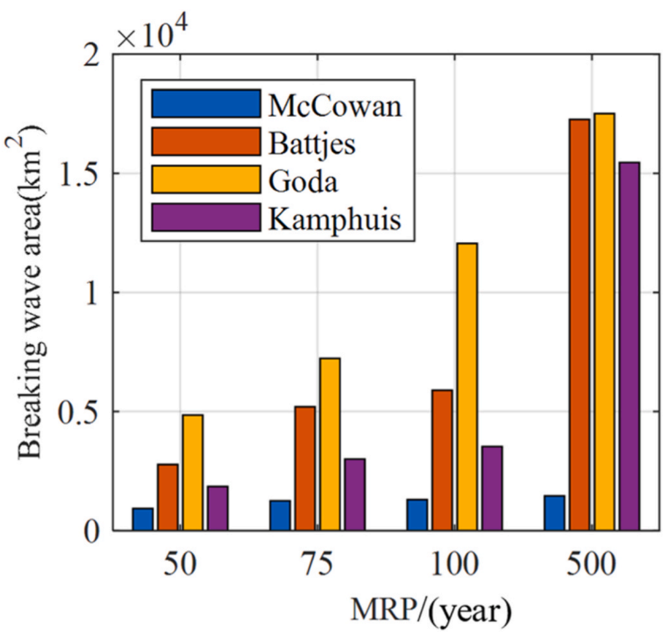

Fig. 17. Breaking wave area of different breaking criteria under different MRPs.

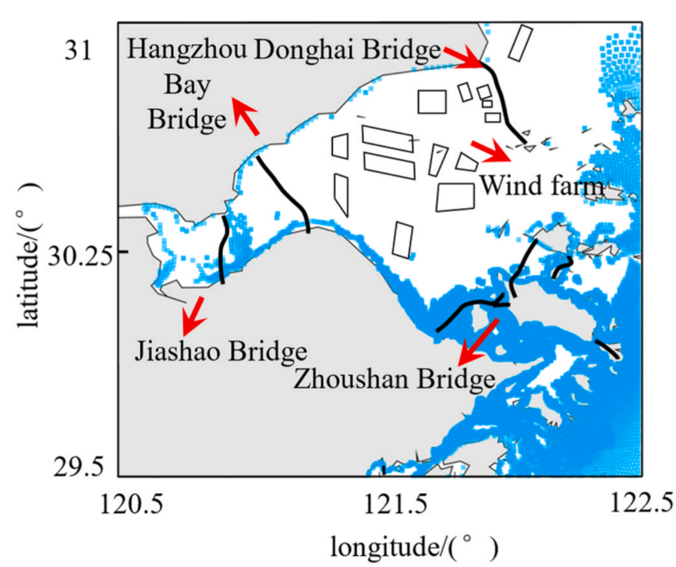

Fig. 18. Breaking wave region of the bivariate probability model under different MRPs.

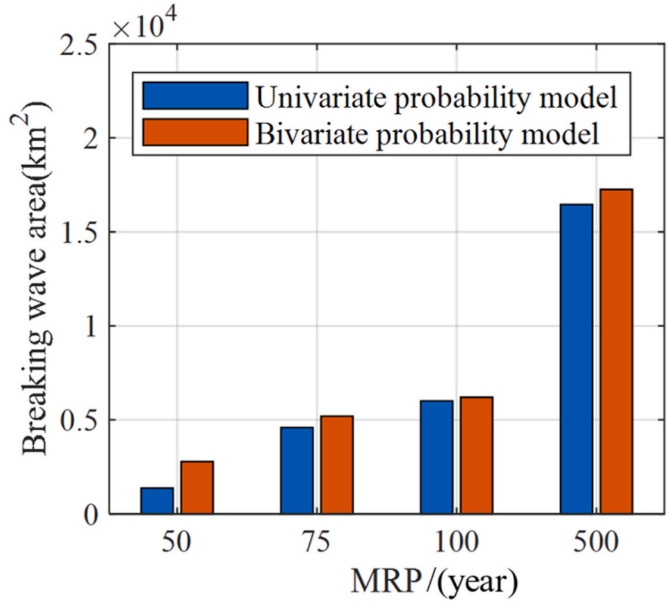

Fig. 19. Comparison of the breaking wave area obtained by univariate and bivariate probability models.

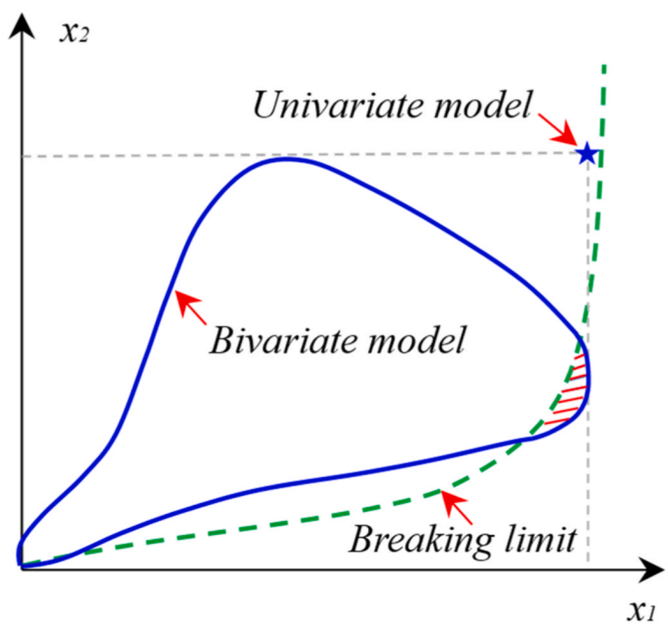

Fig. 20. x_1 and x_2 from univariate and bivariate probability models with the same MRP.

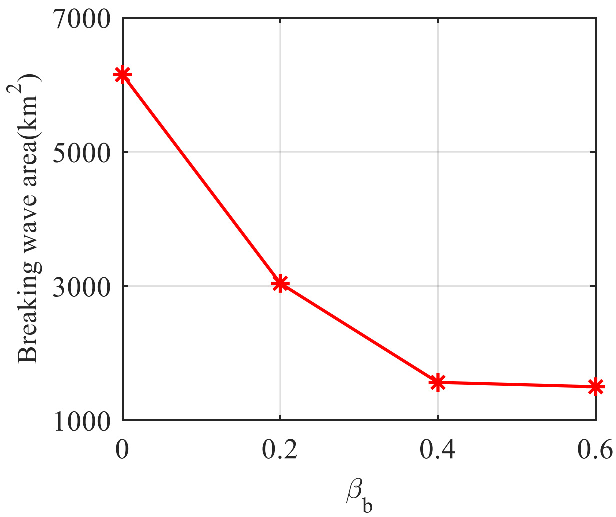

Fig. 22. Comparison of the breaking wave area with different breaking thresholds.

bivariate probability model can only calculate 200 nodes by a computer with a 12-core processor and 32 G memory.

4.3. Effect of breaking threshold for bivariate model

Since breaking wave estimation using a bivariate model depends on the value of the breaking threshold, the breaking wave region with an MRP of 100 years is investigated as a function of the breaking threshold, as shown in Figs. 18 and 21. The spatial distribution and area of the breaking wave region decrease with increasing breaking threshold. When β_b increases from 0 to 0.2, the breaking wave regions in the sea area near the southeast of the Zhoushan Islands are greatly reduced. The breaking wave region on the south coast of Hangzhou Bay and near the coast of the Zhoushan Islands narrows when β_b increases from 0.2 to 0.4. The breaking wave regions have slight changes and are mainly concentrated near the coast when β_b increases from 0.4 to 0.6. There are few bridges affected by breaking waves when considering the effect of the breaking threshold. The breaking wave regions are mainly distributed near the Zhoushan Bridge and occur at the approach bridge near the shore.

As shown in Fig. 22, when the breaking threshold β_b is 0.2, 0.4, and 0.6, the breaking wave area is 3000 km^2, 1560 km^{2,} and 1500 km^2, respectively. The breaking wave area decreases with increasing the breaking threshold β_b for the bivariate probability model. The distribution and area of the breaking wave region depend on the value of the breaking threshold. Therefore, the breaking threshold should be selected properly according to the importance of the offshore structure as well as the associated mitigation measures.

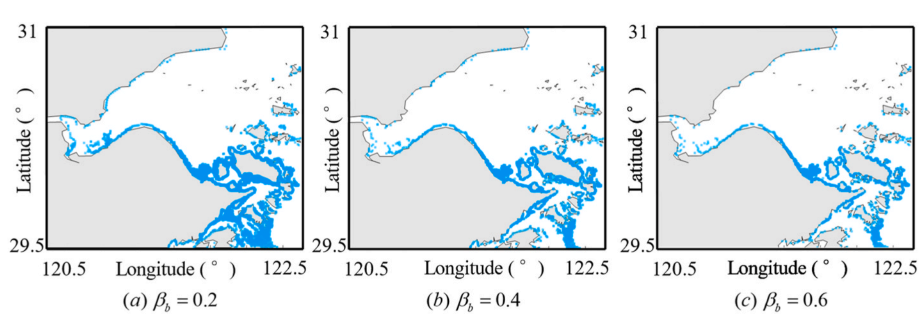

Fig. 21. Breaking wave region with different breaking thresholds under the Battjes criterion.

5. Concluding remarks

This study developed a numerical approach to assess the location and area of breaking wave regions in a given site under typhoons. Taking Hangzhou Bay as an example site, coupled wave and storm surge simulations of 40 historical typhoons from 1987 to 2012 were carried out. The maximum wave and storm surge under each typhoon were obtained to generate the database for breaking wave estimation. The breaking wave regions in Hangzhou Bay under typhoons are analyzed and illustrated based on the simulated database. The effects of breaking criteria, probability type, and breaking threshold on the distribution and breaking wave area were investigated. The major findings can be summarized as follows:

(1) The SWAN + ADCIRC coupling model can provide sufficient wave and storm surge data under historical typhoons, which can be an efficient supplement for sites without long-term measurements. The time lag between the peaks of two breaking prediction variables is not significant and can be neglected.
(2) With the increase in MRP, the breaking wave area increases and expands outward from the coast. The waves on the south shore break more easily than those on the north shore of Hangzhou Bay, but the distribution and area of the estimated breaking wave region depend on the selection of breaking criteria. The breaking wave regions estimated by the Kamphuis, Battjes, and Goda criteria include more factors and agree with the actual events better than the McCowan criterion.
(3) The breaking wave region estimated using the univariate and bivariate probability models are compared. In the estimation of the breaking wave, the univariate model has only one point of data at the given MRP, while the bivariate model is a contour. The bivariate model is more rigorous than the univariate model in judging breaking. Neglecting the correlation between two breaking prediction variables results in a smaller spatial distribution and area of the breaking wave region. The breaking wave area decreases with increasing the breaking threshold for the bivariate probability model. The distribution and area of the breaking wave region depend on the value of the breaking threshold. The breaking threshold should be selected properly according to the importance of the offshore structure as well as the associated mitigation measures.

It should be noted that this study only presented the estimation approach of typhoon-induced breaking waves from the perspective of engineering practice. The estimation precision of the developed approach depends on the simulation accuracy of the SWAN + ADCIRC model. The parametric wind field model can be improved by a more advanced WRF model (Zhong et al., 2023). More field measurements of wave conditions under typhoons should be incorporated to validate the approach. Moreover, the historical tropical cyclones of the example study are not long enough to capture the tail of the wave and storm surge hazards with long MRP. Synthetic typhoon technology can be used to expand the typhoon catalog and include the stochastic nature (Wei et al., 2023). Notwithstanding the limitations, this study provides a powerful numerical tool for breaking wave risk assessment. It will be of particular interest to engineers in determining the design load of typhoon-induced waves.

CRediT authorship contribution statement

Wenyu Zhao: Investigation, Methodology, Data curation, Writing – original draft. **Kai Wei:** Conceptualization, Supervision, Visualization, Writing – review & editing. **Xi Zhong:** Software, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Acknowledgments

The authors would like to acknowledge the financial support of the National Natural Science Foundation of China (Grant Nos. 51978578 and 52222804) and the National Key Research and Development Program of China (Grant No. 2021YFB1600300).

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